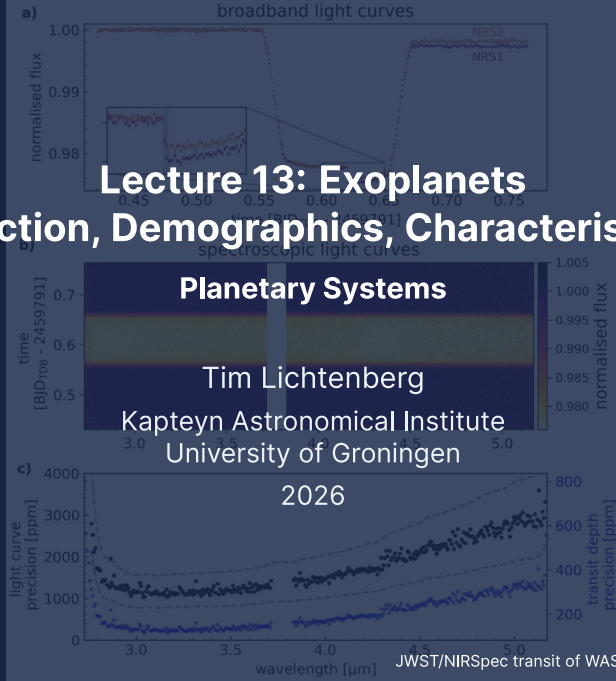


# Lecture 13: Exoplanets

## Detection, Demographics, Characterisation



Planetary Systems

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University of Groningen

## Learning Objectives

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By the end of this lecture, you will be able to:

- Describe the main exoplanet detection methods and their biases
- Combine transit and radial-velocity geometry into a bulk density, and read the radius valley
- Evaluate JWST-era atmospheric results and biosignature claims

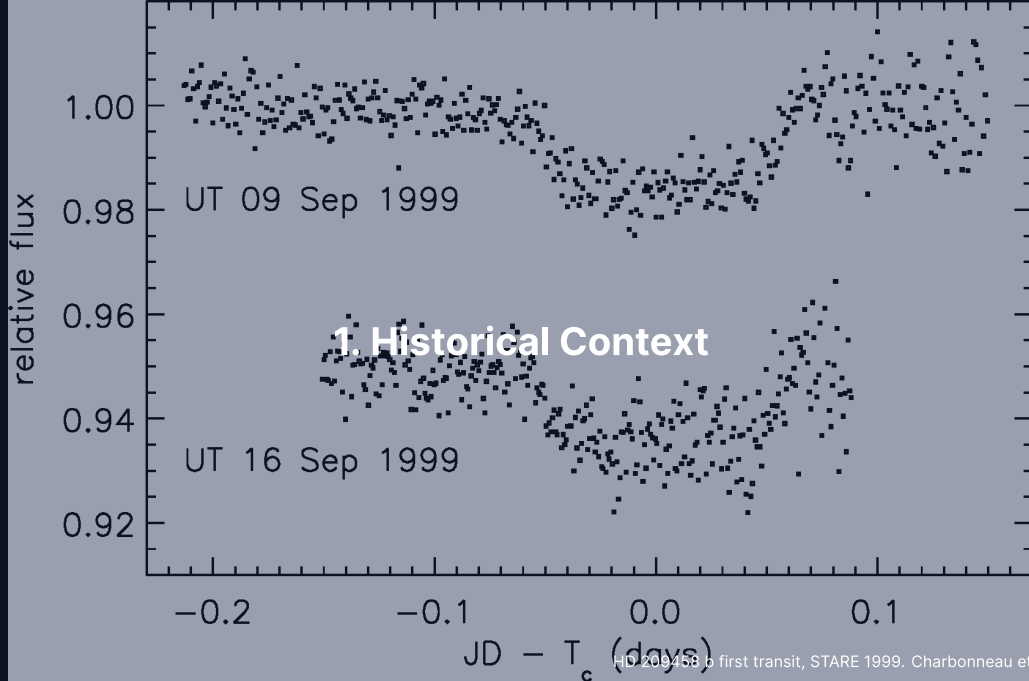
From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

## Recap: Where We Are in the Course

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- Lectures 9–11 surveyed the geology, atmospheres, and interiors of the solar planets
- Lecture 12 decoded small bodies and meteorites as planet formation fossils
- This lecture turns outward to thousands of exoplanets to test if these patterns are universal

**Today:** How do we detect, weigh, and characterise worlds beyond our solar system?



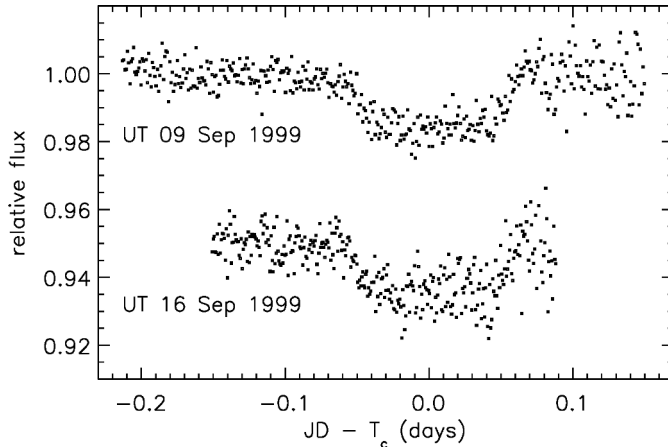
## 1. Historical Context

HD 209458 b first transit, STARE 1999. Charbonneau et al. (2000)

## From Zero to Six Thousand in Three Decades

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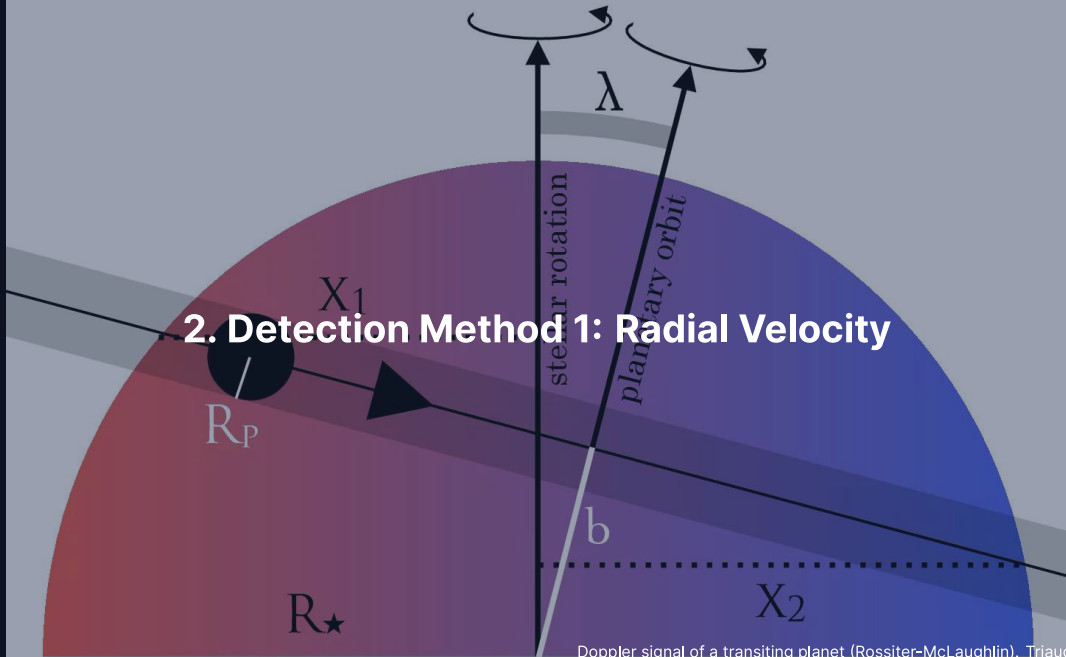
- **1992 (PSR B1257+12):** first exoplanets, around a pulsar, by timing (Wolszczan & Frail)
- **1995 (51 Pegasi b):** first planet around a Sun-like star; a 4.23-day orbit forced migration theory (Mayor & Queloz)
- **1999 (HD 209458 b):** first transit of an RV planet: a gas-giant radius and bulk density



The first ground-based exoplanet transit: HD 209458 b with the STARE photometer on two nights in September 1999. Charbonneau et al. (2000).

Successive transits dropped the flux by about 1.7% (independent detection by Henry et al. 2000): exoplanets became physical objects with measurable radii.

## 2. Detection Method 1: Radial Velocity



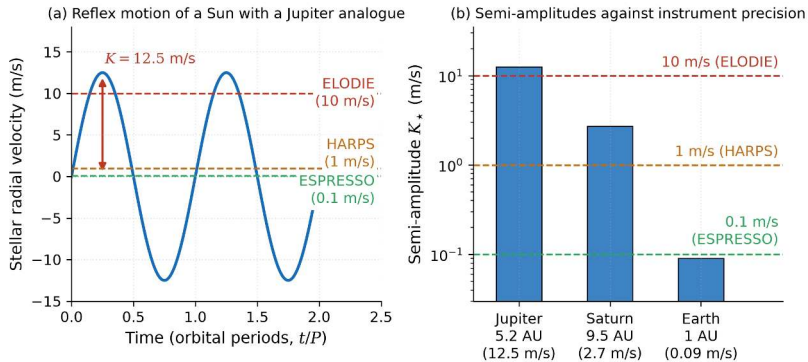
## Radial Velocity: Doppler Reflex of the Star

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- Star and planet orbit a common barycentre:  $a_{\star} = (m_p/M_{\star}) a_p$
- The periodic Doppler shift of the stellar absorption lines gives the reflex velocity
- Semi-amplitude  $K_{\star}$  for inclination  $i$  (edge-on:  $\sin i = 1$ ; circular:  $e = 0$ ):

$$K_{\star} = \left( \frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

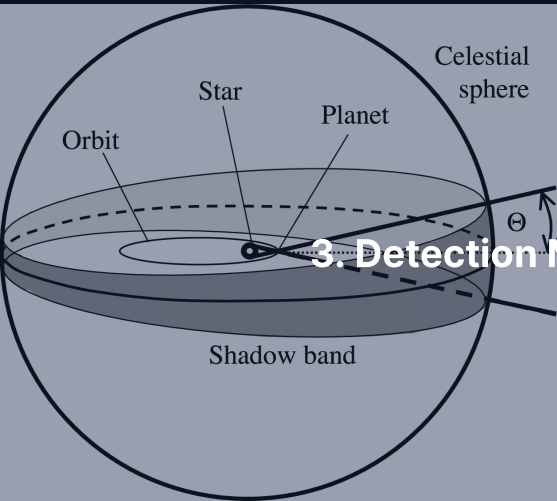
Jupiter at 5.2 AU:  $K_{\star} \approx 12.5 \text{ m s}^{-1}$ ; Earth at 1 AU:  $K_{\star} \approx 9 \text{ cm s}^{-1}$ .



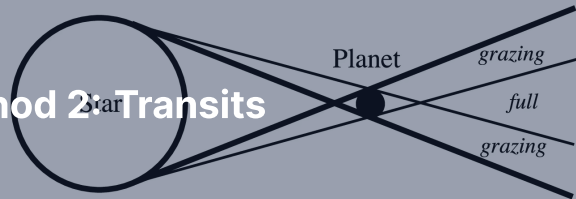
(a) The reflex velocity of a Sun with a Jupiter analogue against the precisions of ELODIE, HARPS and ESPRESSO; (b) the semi-amplitudes of Jupiter, Saturn and Earth analogues.

Course figure.

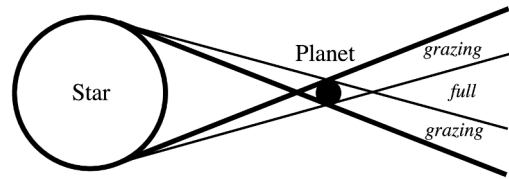
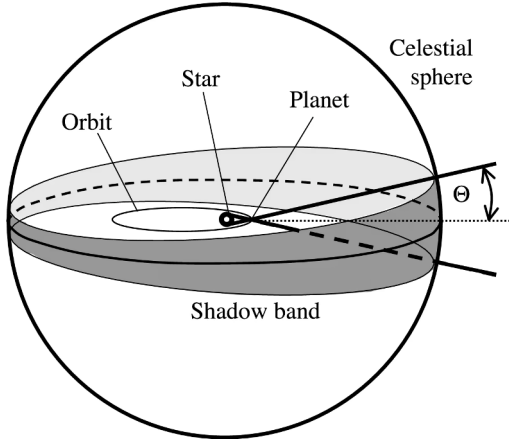
An Earth analogue at  $0.09 \text{ m s}^{-1}$  sits below the instrumental floor and within the  $0.1\text{--}1 \text{ m s}^{-1}$  stellar jitter; RV alone gives only the minimum mass  $m_p \sin i$ .



### 3. Detection Method 2: Transits



Close-up



Close-up

Transit geometry: a planet whose orbital plane lies near the line of sight crosses the stellar disk. Winn (2010).

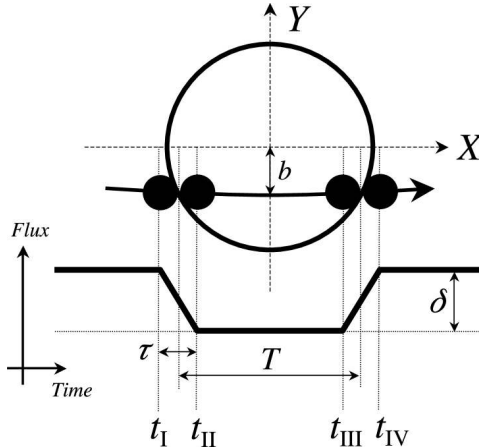
Transit depth  $\delta = (R_p/R_\star)^2$  and geometric probability  $p \approx R_\star/a$ : a strong selection bias toward short periods and large  $R_p/R_\star$ .

## Quick Scales

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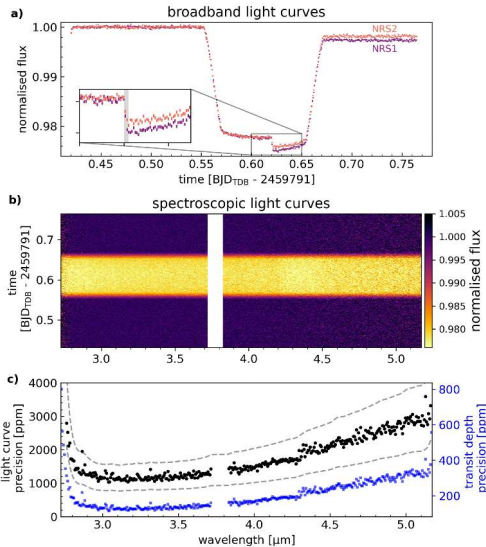
Transit depth scales inversely with the square of the stellar radius, so small planets around small stars are far easier to detect.

- **Jupiter around Sun:**  $\delta \approx 1.01\%$ , easily detected from ground-based telescopes
- **Earth around Sun:**  $\delta \approx 84$  ppm, requires ultra-precise space photometry
- **Earth around M dwarf ( $0.15 R_{\odot}$ ):**  $\delta \approx 4000$  ppm, accessible from ground and space



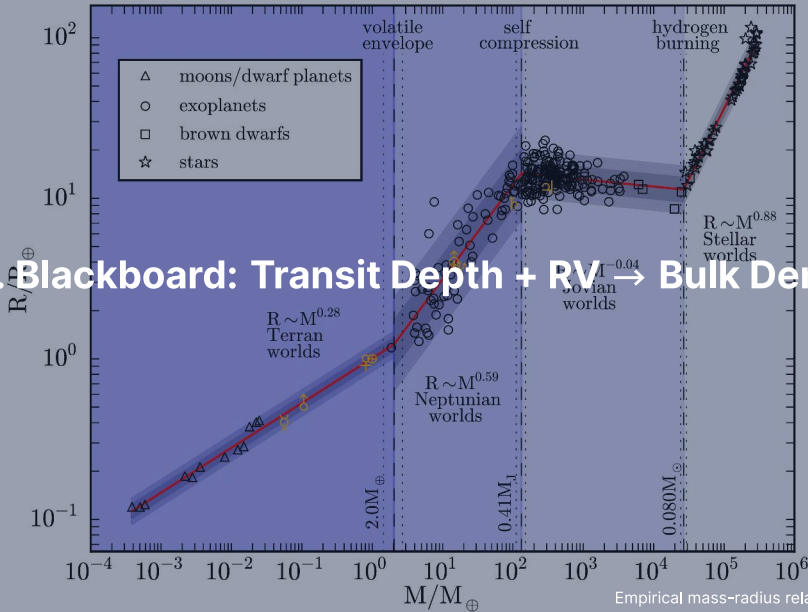
Anatomy of a transit light curve with its four contact times  $t_I$  to  $t_{IV}$ . Winn (2010).

The depth gives the radius ratio  $R_p/R_*$ ; the total duration and the ingress time give the impact parameter  $b$  and the inclination. A real transit bottom curves from stellar limb darkening.

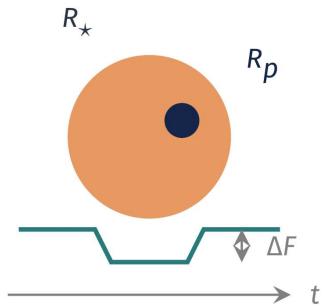


JWST/NIRSpec transit photometry of WASP-39 b: depths measured to 200–600 ppm per spectroscopic bin, below typical exoplanet absorption features. Alderson et al. (2023).

# 4. Blackboard: Transit Depth + RV $\rightarrow$ Bulk Density



## Goal and strategy



**Goal:** a radius and a mass for the same planet, hence a density.

- Transit alone: the radius  $R_p$ , but no mass
- RV alone: the minimum mass  $m_p \sin i$ , but no radius
- Combined: **both**, so the bulk density  $\bar{\rho}_p$

Bulk density turns an astronomical detection into a physical, geological body. To the board.

## Result: Depth, Semi-Amplitude, Density

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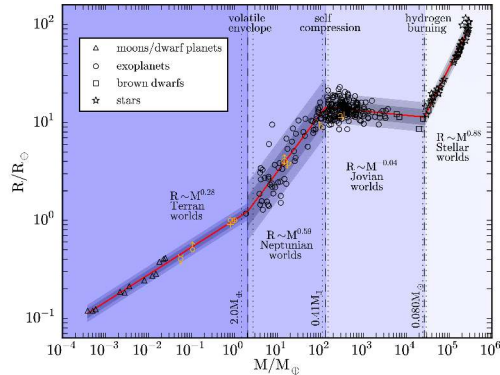
Transit depth  $\delta = (R_p/R_\star)^2$  gives  $R_p$ ; the reflex velocity

$K_\star = (2\pi G/P)^{1/3} m_p \sin i M_\star^{-2/3}$  gives  $m_p \sin i$  (broad).

- A transit forces  $\sin i \approx 1$  and collapses the  $m_p \sin i$  degeneracy
- Both  $R_p$  and  $m_p$  measured for the same object
- Bulk density follows from elementary geometry:

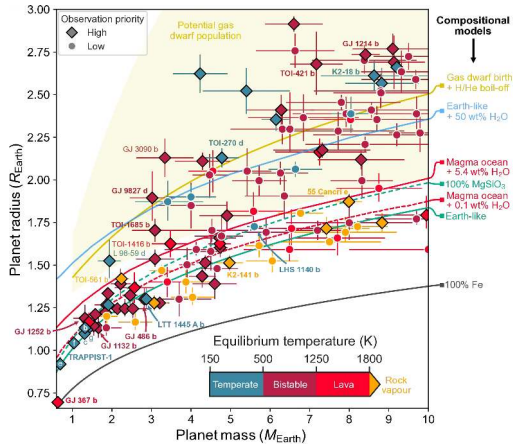
$$\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}$$

Transit plus RV turns a detection into a physical object: radius, mass and density from one planet.



The empirical mass-radius relation of exoplanets and solar-system bodies. Chen & Kipping (2017).

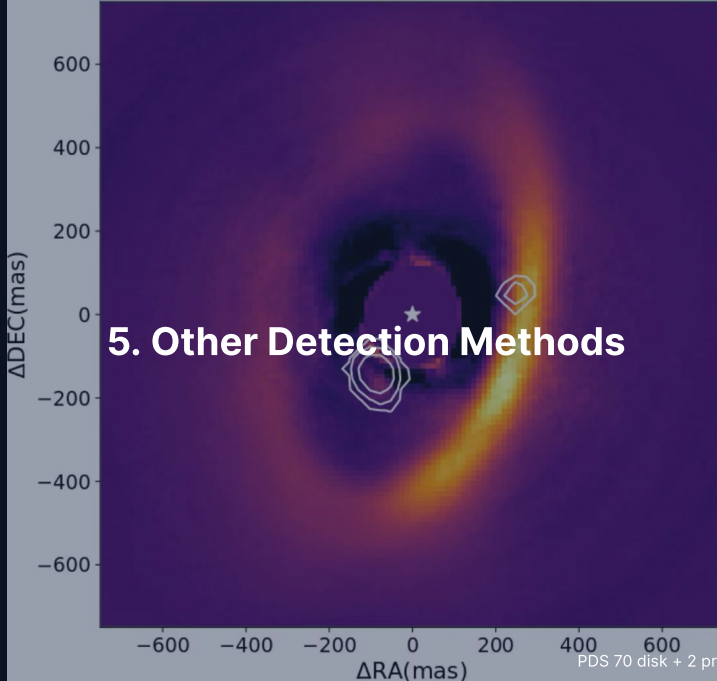
Density maps to composition:  $\bar{\rho} \approx 5.5 \text{ g cm}^{-3}$  rocky (Earth, Venus),  $1.3 \text{ g cm}^{-3}$  gas giant (Jupiter),  $0.5 \text{ g cm}^{-3}$  an inflated envelope, and  $1\text{--}3 \text{ g cm}^{-3}$  the sub-Neptunes, a size class the solar system lacks.



Mass against radius for small exoplanets with interior models from pure iron to volatile-rich envelopes, sorted by thermal regime. Reproduced from Lichtenberg et al. (2025).

Rocky super-Earths follow the silicate-iron curves; sub-Neptunes need substantial volatile envelopes or water-rich layers to explain their larger radii.

## 5. Other Detection Methods

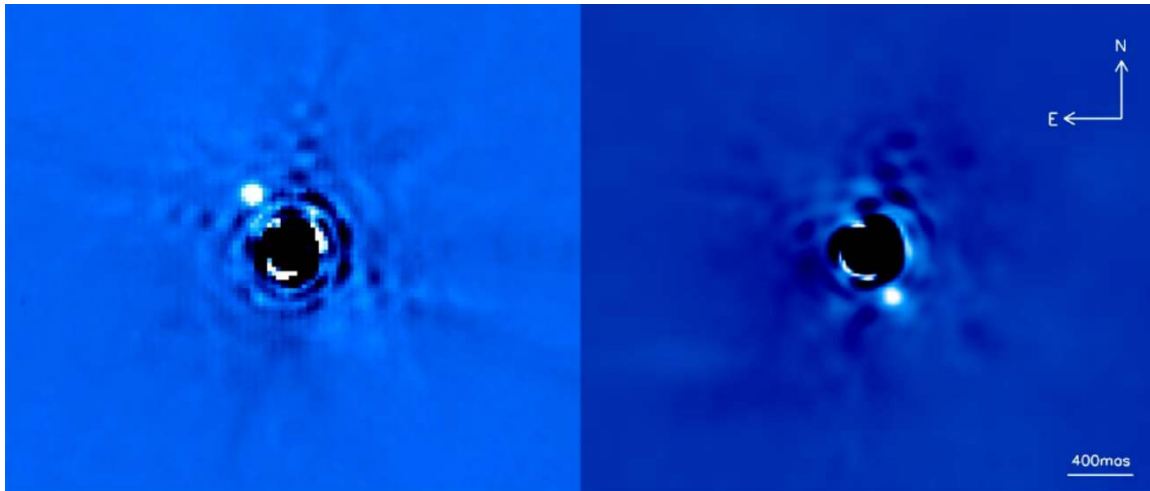


PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

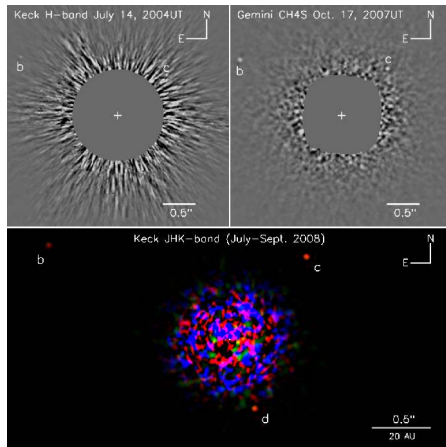
## Direct Imaging: Spatial Separation of Photons

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- **Contrast:** suppress the star by  $10^{-9}$  (Jupiter) to  $10^{-10}$  (Earth twin) at separation  $\theta = a/d$
- **Technology:** extreme adaptive optics plus coronagraphs against diffraction and speckle noise
- **Bias:** young, massive, self-luminous giants on wide orbits ( $a > 10$  AU)

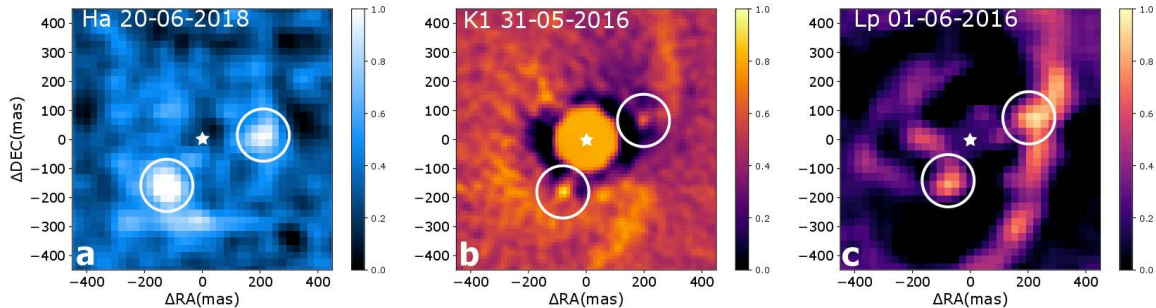


Direct image of  $\beta$  Pictoris b: orbital motion between 2003 and 2009 confirms a bound  $\sim 9 M_J$  companion at  $\sim 9$  AU. Lagrange et al. (2010); credit: ESO/A.-M. Lagrange et al.

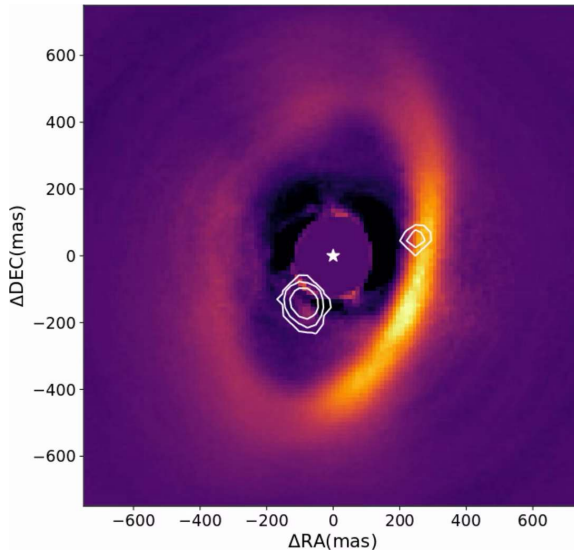


Keck and Gemini adaptive-optics discovery images of the HR 8799 planets. Marois et al. (2008).

Four co-moving planets of 5–10  $M_J$  at 14–68 AU (planet e added in 2010) around a 30–60 Myr star: they still glow from their contraction heat.



PDS 70 b and c inside the disk gap in MUSE H $\alpha$  (a), SPHERE K1 (b) and NACO L' (c): both protoplanets are accreting within the cleared cavity. Haffert et al. (2019).



Composite scattered-light image of the PDS 70 protoplanetary disk with the two embedded protoplanets inside the cleared gap. Credit: Haffert et al. (2019).

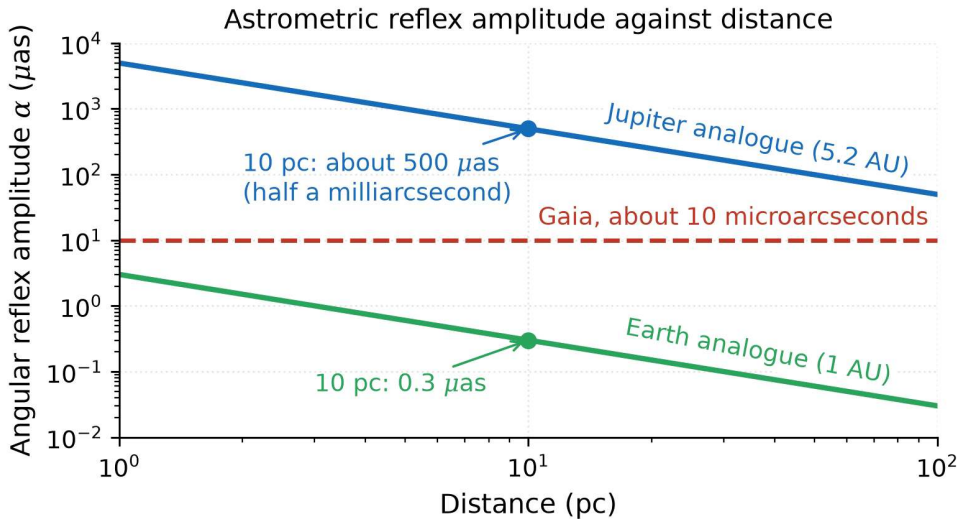
## Astrometry: Gaia Will Find Jupiter Analogues

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Astrometry follows the star's reflex wobble on the sky and gives true masses.

$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- **Signal:** a Jupiter analogue at 10 pc gives  $\alpha \sim 500 \mu\text{as}$ ; an Earth analogue  $\sim 0.3 \mu\text{as}$
- **Gaia DR4 (2026):** thousands of cold giants expected;  $\sim 10 \mu\text{as}$  final precision
- **True mass:** the inclination  $i$  is measured, so no  $m \sin i$  degeneracy



Astrometric reflex amplitude against distance for Jupiter and Earth analogues, with Gaia's final precision of about 10 microarcseconds as the dashed line. Course figure.

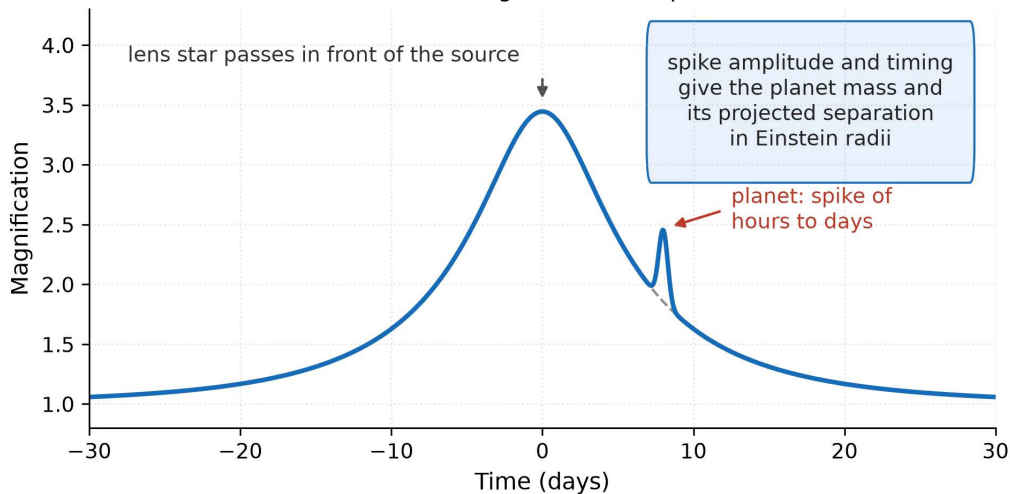
## Microlensing, Timing, and Detection Biases

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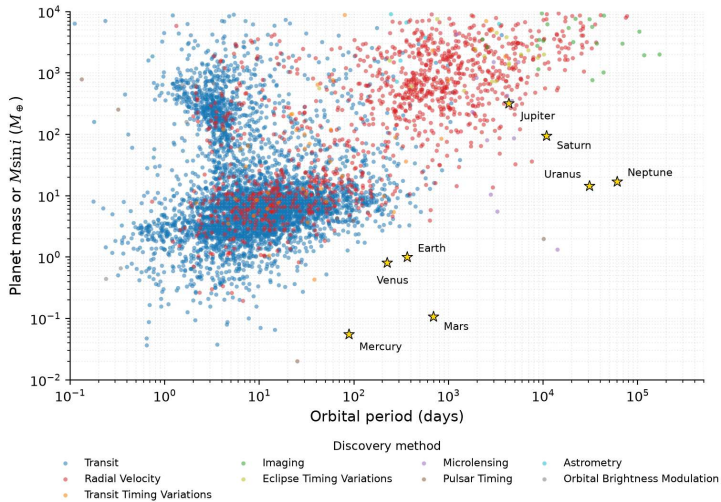
- **Microlensing (Roman):** cool planets at 1–10 AU and unbound rogue worlds
- **Timing variations (TTVs):** dynamical masses for faint hosts such as TRAPPIST-1
- **Together:** transits and RV probe close-in orbits; imaging, astrometry and microlensing the wide ones

Each method covers its own part of parameter space, so the archive carries the biases of the methods.

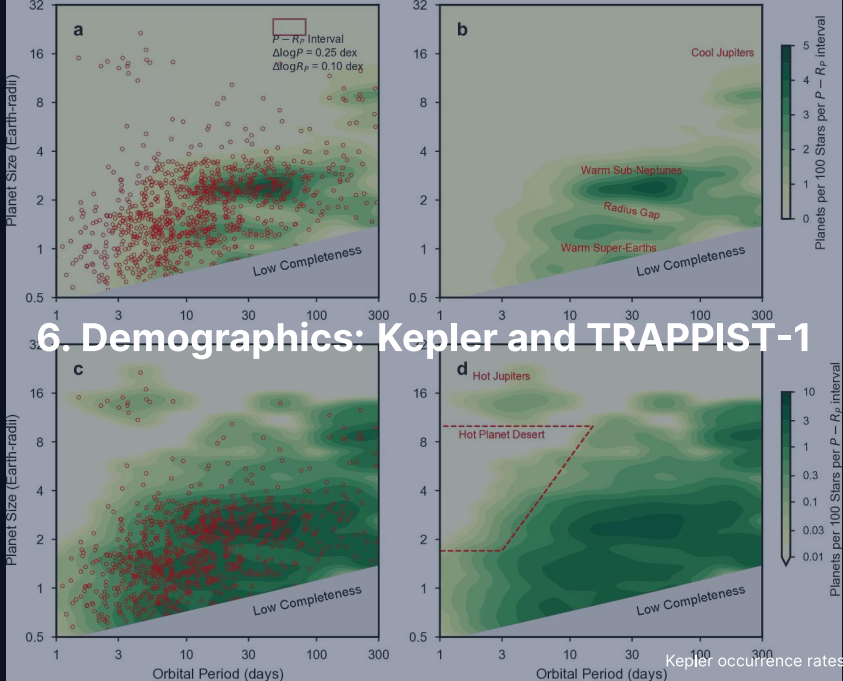
## A microlensing event with a planet



A microlensing event with a planet: the magnification of a background source by a lens star, and the spike of hours to days that a planet around the lens adds. Course figure.



Confirmed exoplanet masses (or  $m_p \sin i$ ) against orbital period by detection technique, with the solar-system planets: discoveries pile up at high mass and short period. Credit: NASA Exoplanet Archive (2026). Course figure.



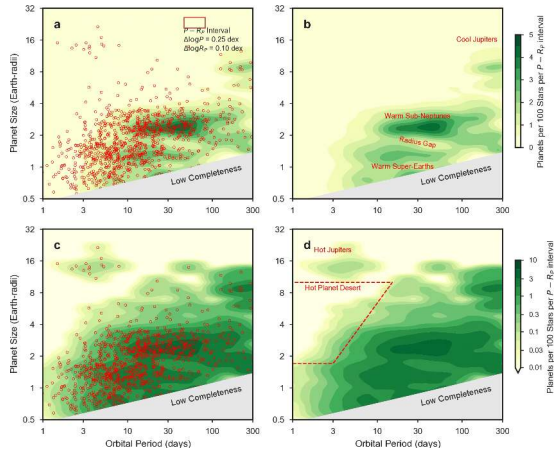
## 6. Demographics: Kepler and TRAPPIST-1

# The Kepler Revolution

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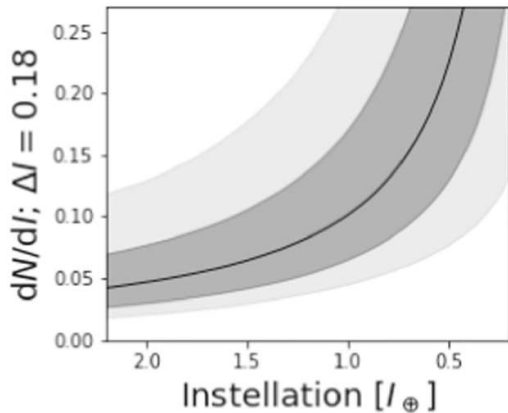
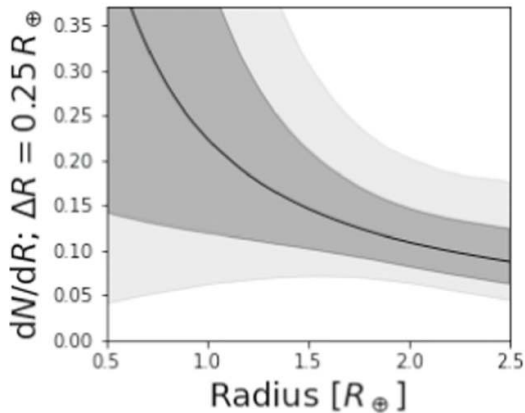
- **Planets are ubiquitous:** on average more than 1 planet per main-sequence star (Petigura et al. 2018)
- **Small planets dominate:** super-Earths and sub-Neptunes are common; hot Jupiters are rare (0.5–1%)
- **Habitable terrestrial rate:** Earth-sized planets in the habitable zone occur at  $\eta_{\oplus} \sim 0.4$  (Bryson et al. 2021)

Kepler's computable detection efficiency gave the first bias-corrected occurrence rates: planets outnumber stars in the Milky Way.



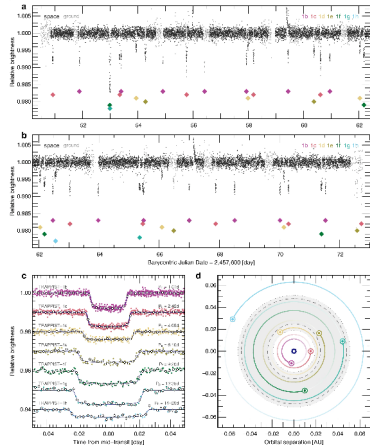
Planet occurrence against orbital period and size from the California-Kepler Survey after stellar parameter refinement. Petigura et al. (2018).

Small planets outnumber giants at every period; the fall-off beyond ~ 100 days is partly completeness-limited.



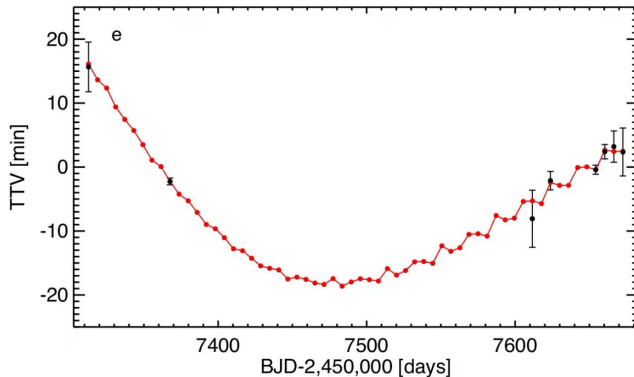
The occurrence rate of Earth-sized planets in the habitable zones of Sun-like stars, marginalised over radius and instellation, with 68% and 95% credible bands. Bryson et al. (2021).

The central estimate  $\eta_\oplus \sim 0.4$ : terrestrial planets in habitable zones are common.



Transits of the seven TRAPPIST-1 planets. Gillon et al. (2017).

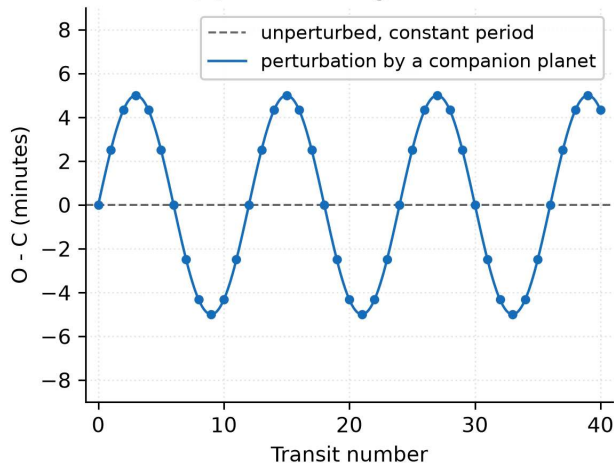
An M8 ultra-cool dwarf at 12 pc with seven Earth-sized planets within about 0.06 AU in a resonant chain, consistent with early inward migration; e, f and g orbit in the temperate habitable zone.



Transit-timing variations of TRAPPIST-1 e: observed deviations against a dynamical model coupled to the other six planets. Gillon et al. (2017).

Timing variations of tens of minutes, fitted with simultaneous N-body models, break the mass degeneracies and give individual masses and densities without radial velocities.

(a) Transit timing variations



(b) Timing methods

**transit timing variations:**

dynamical masses (TRAPPIST-1)

**pulsar timing:**

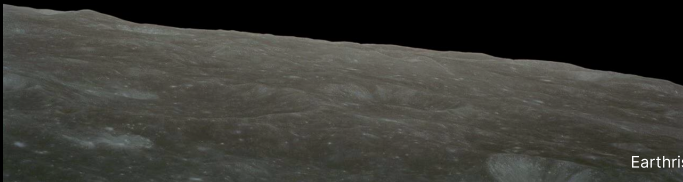
the first exoplanets (1992)

**eclipse timing in binaries:**

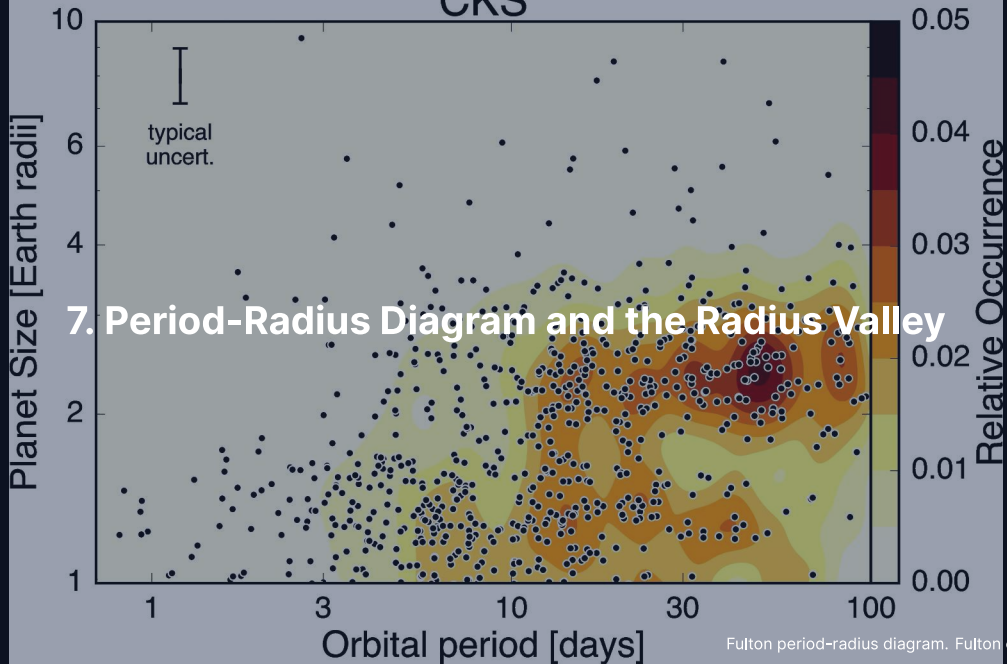
circumbinary planets  
(Kepler-16 b, 2011)

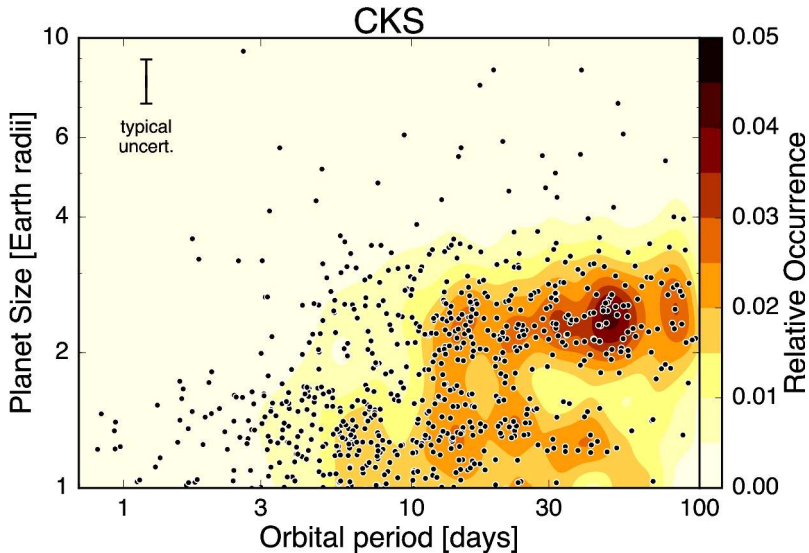
(a) Observed minus calculated transit times of a planet perturbed by a companion (5 minute amplitude, schematic); (b) the three timing methods: transit timing, pulsar timing and eclipse timing in binaries. Course figure.

**Break**

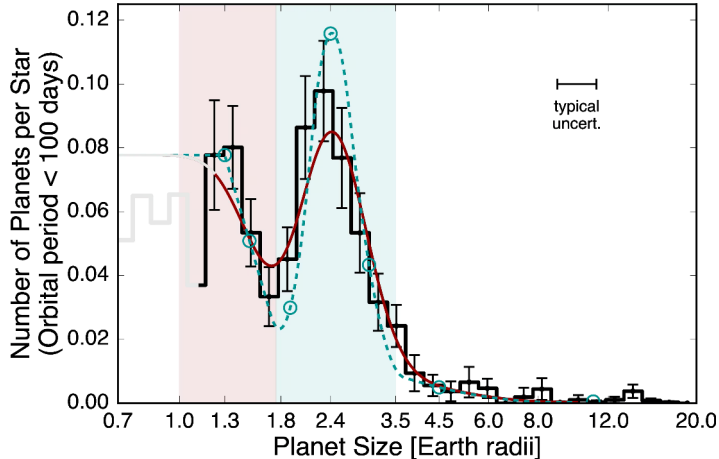


CKS





Period-radius distribution of small Kepler planets after stellar parameter refinement: super-Earths and sub-Neptunes separated by a deficit at  $\sim 1.8 R_{\oplus}$ . Fulton et al. (2017).



The radius valley: the size distribution of small Kepler planets is bimodal. Fulton et al. (2017).

A deficit at  $1.5\text{--}2.0 R_{\oplus}$  separates super-Earths near  $1.3 R_{\oplus}$  (bare rocky cores) from sub-Neptunes near  $2.4 R_{\oplus}$  (volatile envelopes); photoevaporation or core-powered mass loss carves it.

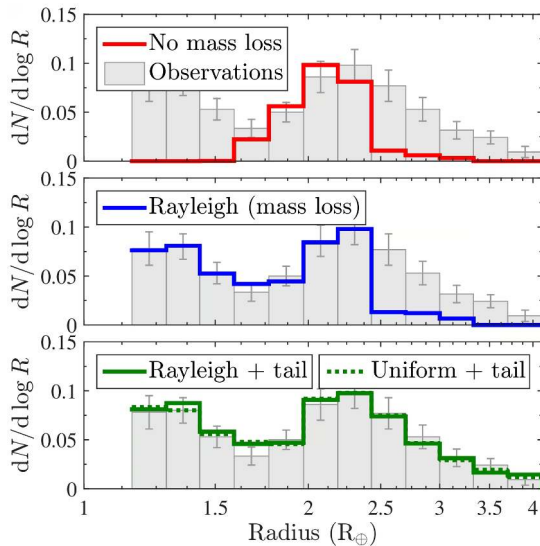
## Photoevaporation Strips H/He Envelopes

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Energy-limited hydrodynamic escape driven by high stellar X-ray and EUV irradiation:

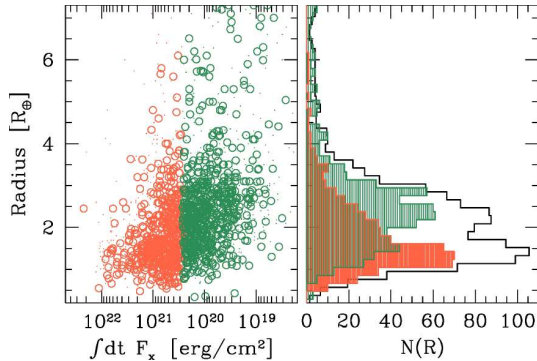
$$\dot{M} \approx \frac{\epsilon \pi F_{\text{XUV}} R_p^3}{GM_p}$$

- Young stars saturate XUV flux at  $\sim 10^3 \times$  modern levels for  $\sim 100$  Myr
- Strips light primordial H/He envelopes from lower-mass cores at close-in orbits
- Owen & Wu (2013): naturally carves the bimodal radius valley

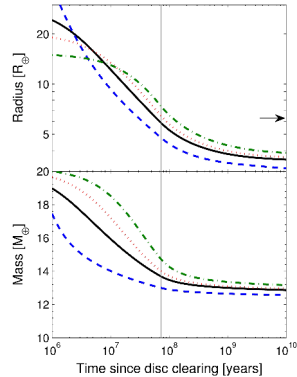


Core-powered mass loss: envelope loss driven by heat from the cooling rocky interior also produces a bimodal distribution with a gap at  $\sim 1.8 R_{\oplus}$ . Credit: Ginzburg et al. (2018).

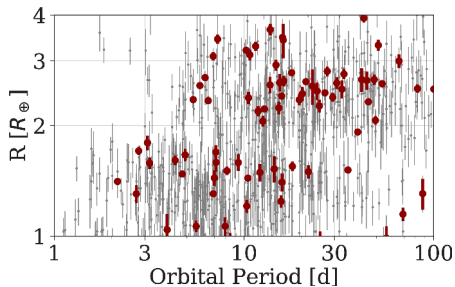
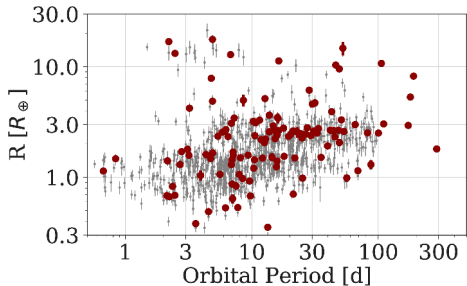
# Photoevaporation Model + Mass-Loss Tracks



Left: predicted radius valley

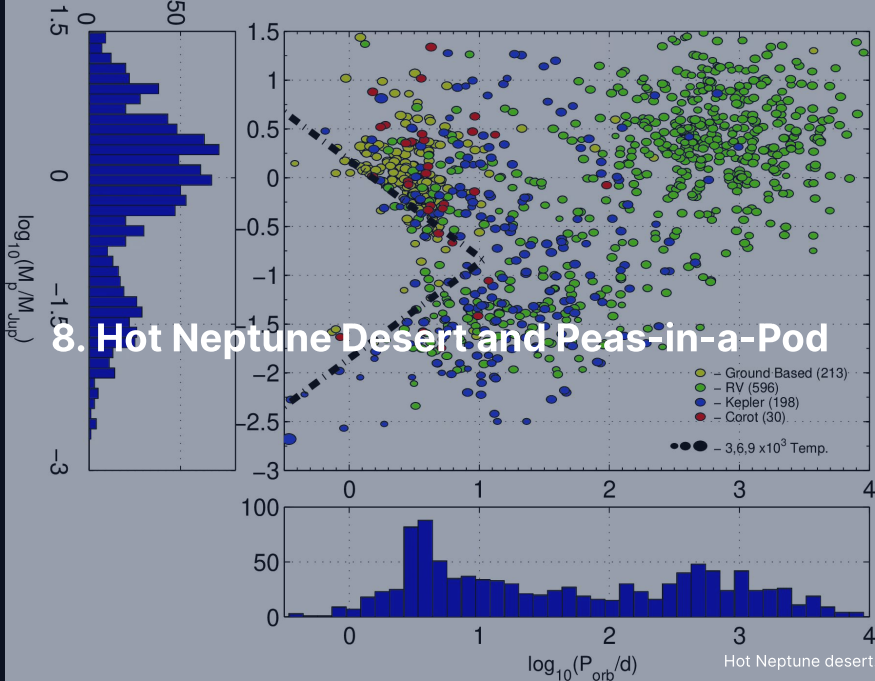


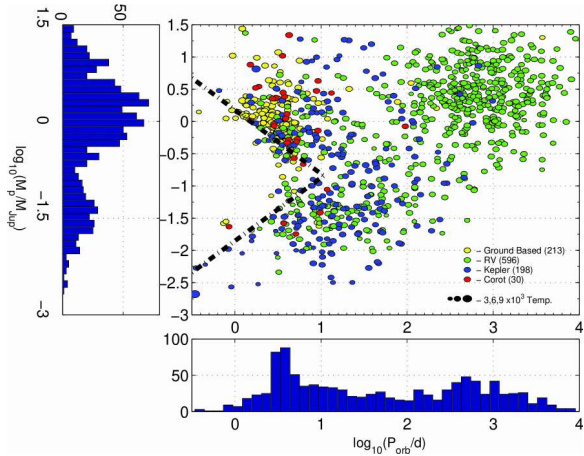
Right: time evolution of radius and mass; the saturated XUV phase drives the bimodal outcome (Owen & Wu 2013).



The radius valley with asteroseismic stellar radii: its position shifts with orbital period. Van Eylen et al. (2018).

The valley slopes to smaller  $R_p$  at longer periods, as photoevaporation and core-powered cooling predict: many close-in super-Earths are stripped cores rather than primordial rocky worlds.

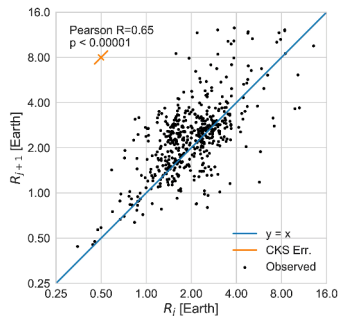




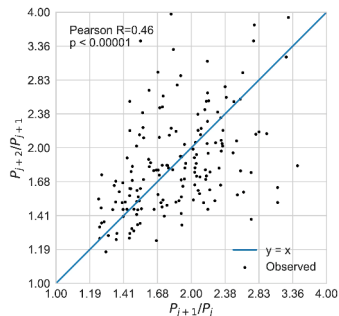
The hot Neptune desert in the mass-period plane. Mazeh et al. (2016).

A deficit of  $10\text{--}100 M_{\oplus}$  planets at  $P < 5$  d: at the **bottom edge** photoevaporation strips low-mass envelopes and leaves super-Earths, at the **top edge** tidal Roche-lobe overflow strips inflated giants.

# Peas-in-a-Pod Architectures



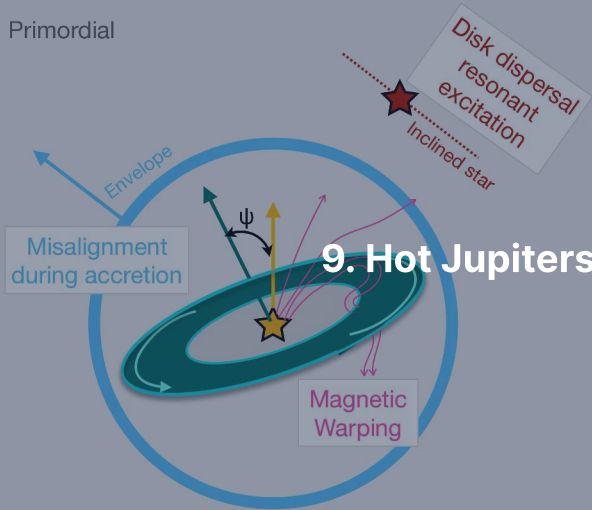
Adjacent radii: Pearson  $r = 0.65$



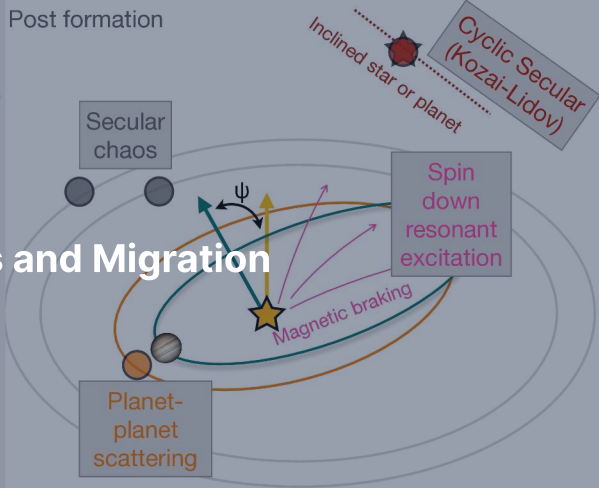
Adjacent period ratios: clustered

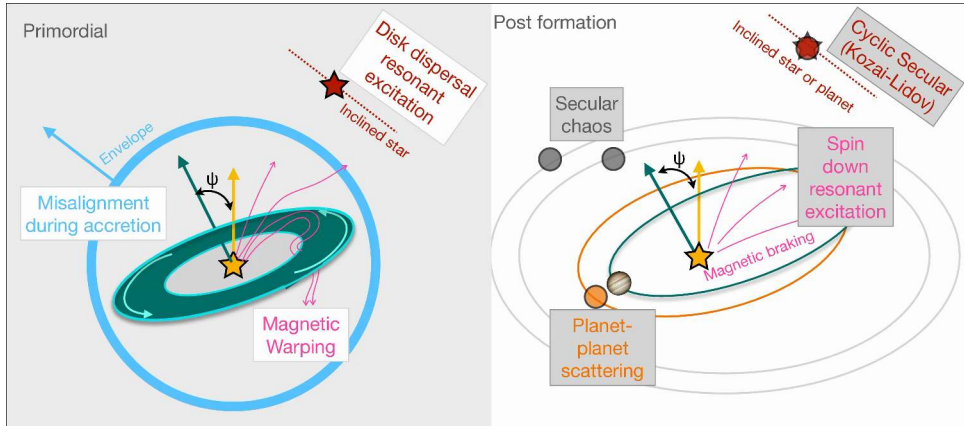
- Planets in compact multi-systems tend to have similar sizes and uniform orbital spacing (Weiss et al. 2018)
- Favours quiet in-situ growth or migration over chaotic giant impacts

Primordial



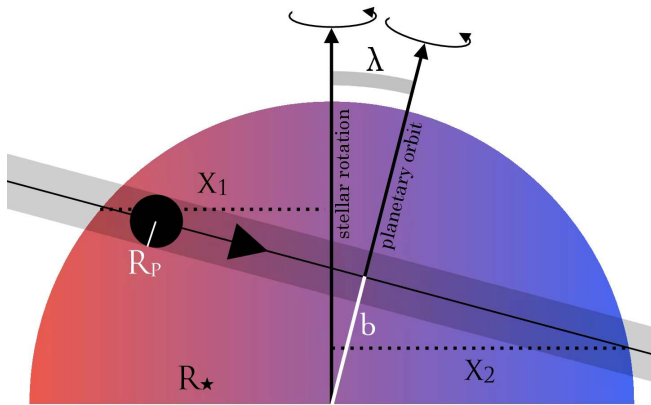
Post formation





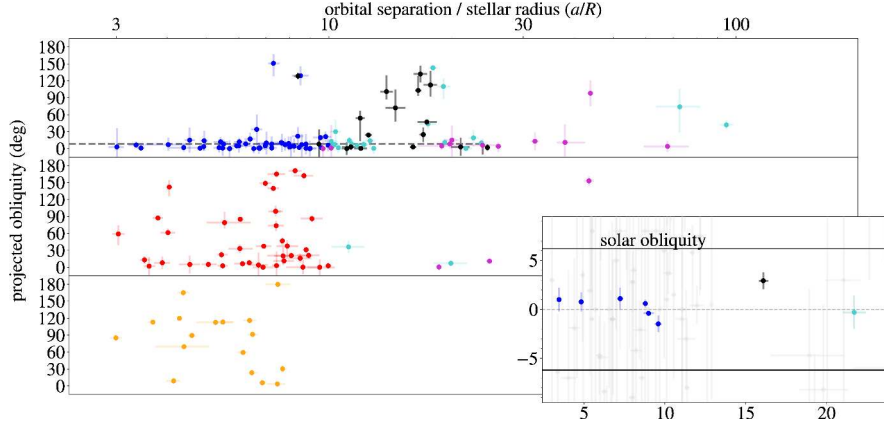
Three pathways to a hot Jupiter and the stellar obliquity each one leaves. Albrecht et al. (2022).

**Disk migration** (Type II) keeps a low obliquity; **high-eccentricity** Kozai-Lidov migration and **planet scattering** leave wide misalignments; in situ growth at 0.05 AU is disfavoured.



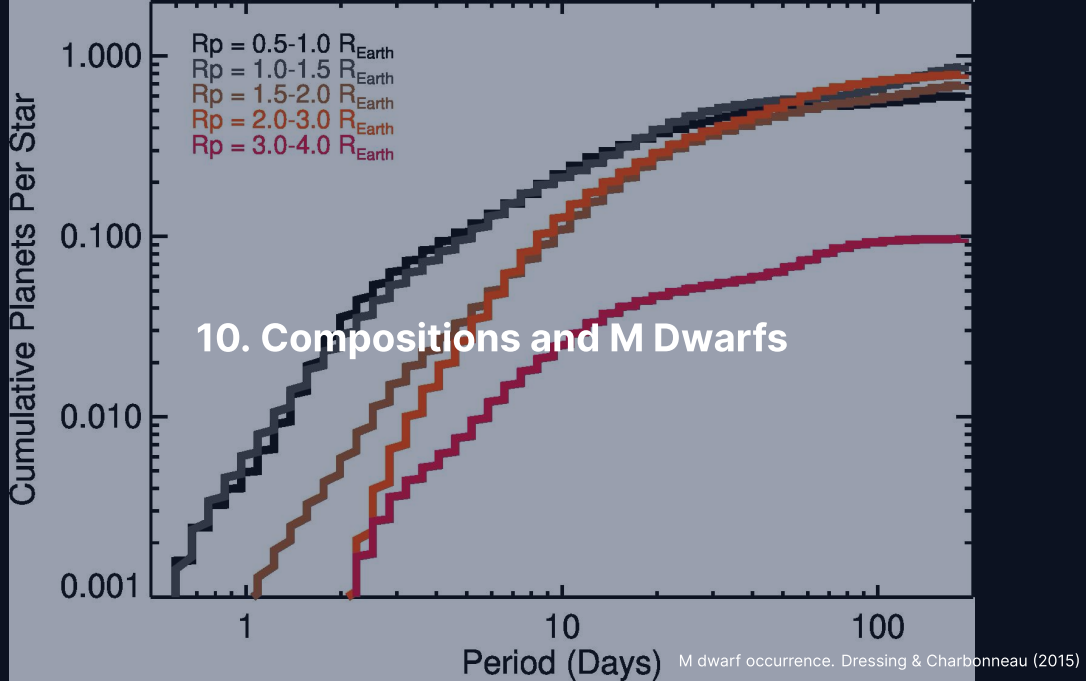
The Rossiter-McLaughlin effect: the Doppler signature of a planet crossing a rotating star.  
Triaud (2018).

On an aligned orbit the planet first hides the approaching, then the receding hemisphere; the time-resolved line distortion gives the sky-projected spin-orbit angle  $\lambda$ , which separates disk migration from dynamical scattering.



Observed stellar obliquities of hot Jupiter hosts against  $a/R_*$ , grouped by stellar effective temperature. Albrecht et al. (2022).

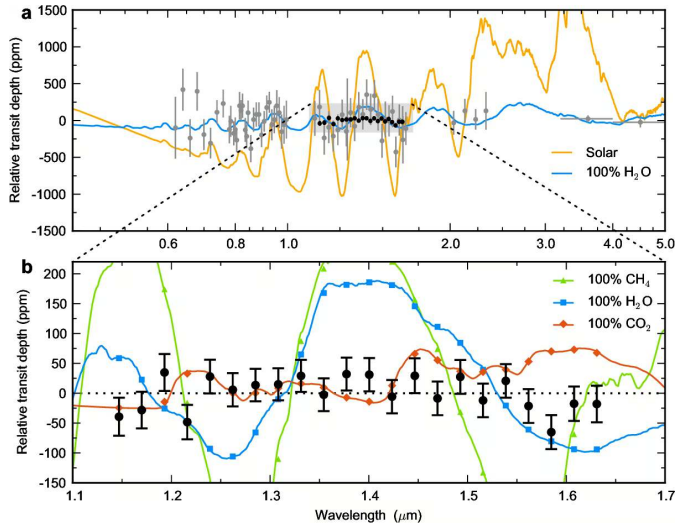
**Cool stars** ( $T_{\text{eff}} < 6250$  K) realign tidally through their convective envelopes; **hot stars** keep high obliquities; the wide dispersion at large  $a/R_*$  shows that dynamical migration occurs.



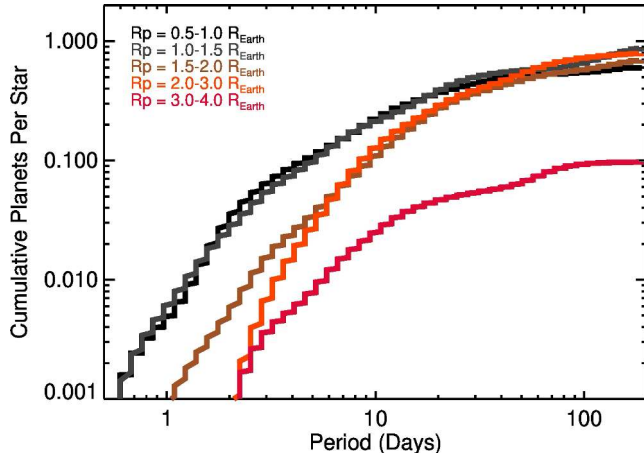
## Sub-Neptunes: Water World or H/He?

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- **Density split:** super-Earths ( $\bar{\rho} \sim 4\text{--}8 \text{ g cm}^{-3}$ ) are rocky; sub-Neptunes ( $\bar{\rho} \sim 1\text{--}3 \text{ g cm}^{-3}$ ) require volatile envelopes
- **Degeneracy:** density cannot separate a 1% H/He envelope from a 50% water-rich mantle
- **Spectroscopic test:** transmission spectroscopy gives the atmospheric mean molecular weight

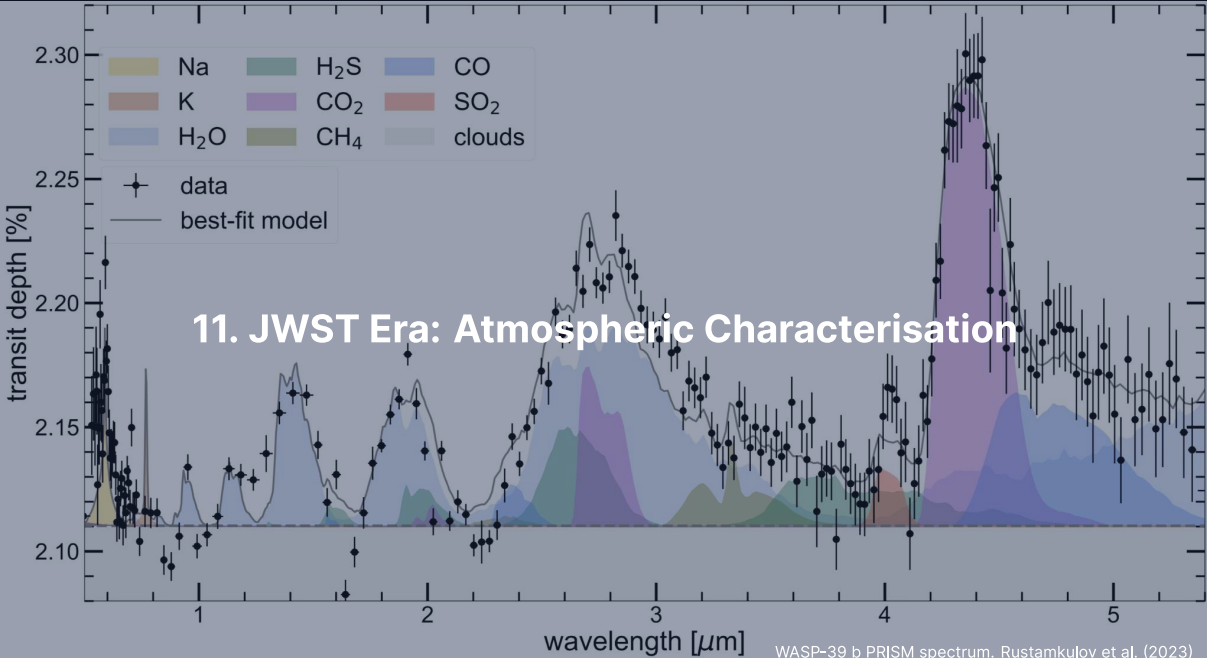


Featureless HST transmission spectrum of the warm sub-Neptune GJ 1214 b: cloud-free  $\text{H}_2\text{O}$ ,  $\text{CH}_4$  and  $\text{CO}_2$  atmospheres are ruled out, so high clouds or hazes are present. Credit: Kreidberg et al. (2014).



Planet occurrence around M dwarfs from Kepler. Dressing & Charbonneau (2015).

M dwarfs host ~ 2.5 small planets each and favour transits and RV, but **XUV desiccation** in the long pre-main sequence can strip whole oceans (Luger & Barnes 2015) and **tidal locking** fixes day and night sides.



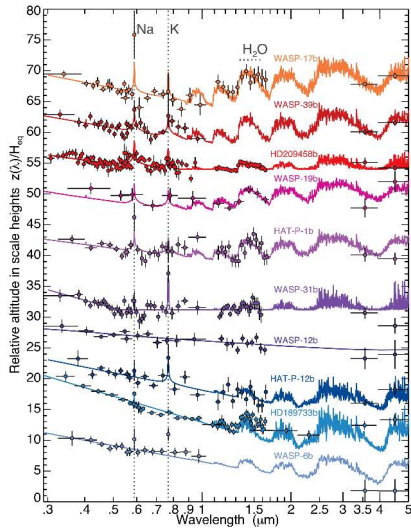
# Transmission Spectroscopy

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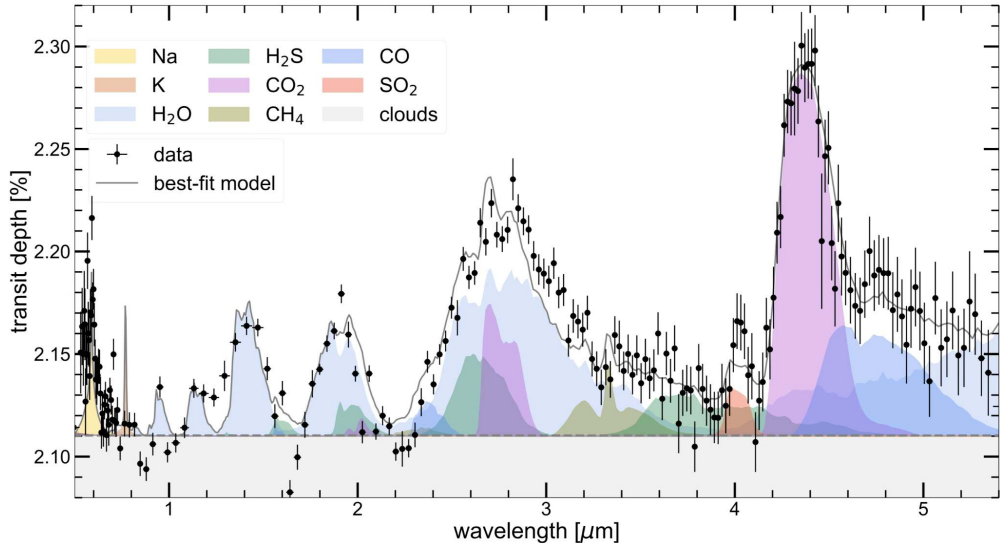
Starlight filtered through an atmosphere during transit encodes molecular absorption; with the scale height

$$H = \frac{k_B T}{\mu m_u g}, \quad \delta(\lambda) \approx \frac{(R_p + n_H(\lambda) H)^2}{R_\star^2}, \quad \frac{\Delta\delta}{\delta} \approx \frac{2n_H H}{R_p}.$$

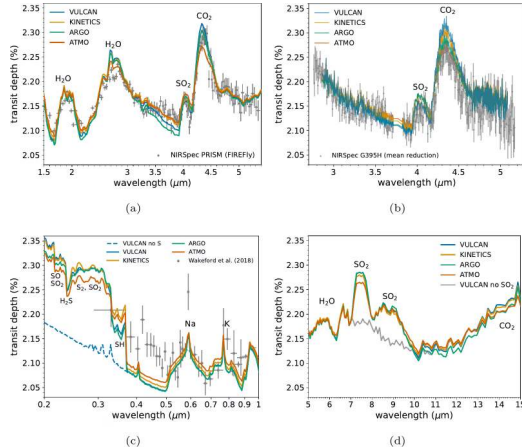
- Hot Jupiter ( $T = 1500$  K,  $\mu = 2.3$ ,  $g = 25 \text{ m s}^{-2}$ ):  $H \approx 200$  km,  $\Delta\delta/\delta \approx 2\%$ , absolute modulation  $\Delta\delta \sim 200$  ppm (JWST-routine)
- Sub-Neptunes: 10–100 ppm; terrestrial atmospheres:  $< 10$  ppm
- High clouds or hazes flatten the spectrum (GJ 1214 b): a flat spectrum does not mean no atmosphere



Transmission spectra of ten hot Jupiters from HST and Spitzer across 0.3–5  $\mu\text{m}$ : a continuum from clear atmospheres with water absorption to cloudy scattering slopes. Credit: Sing et al. (2016).

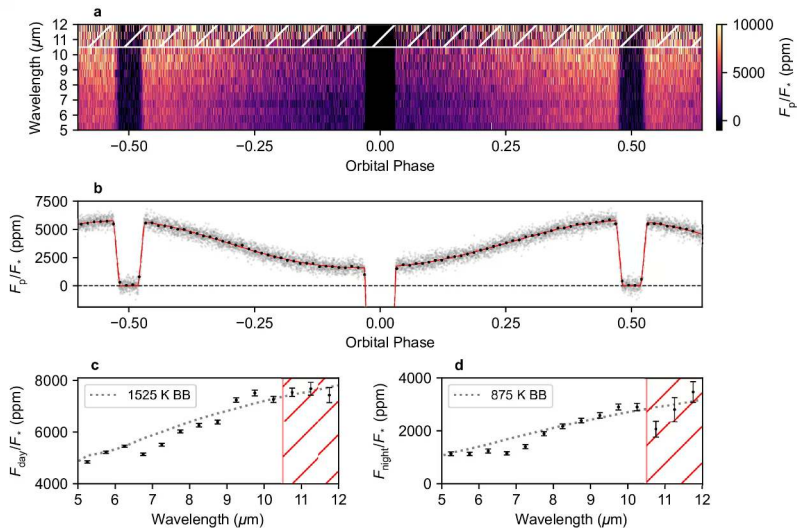


JWST/NIRSpec PRISM transmission spectrum of WASP-39 b with model opacity bands:  $\text{H}_2\text{O}$  ( $33\sigma$ ),  $\text{CO}_2$  ( $28\sigma$ ), clouds ( $21\sigma$ ), Na ( $19\sigma$ ) and CO ( $7\sigma$ ). Rustamkulov et al. (2023).



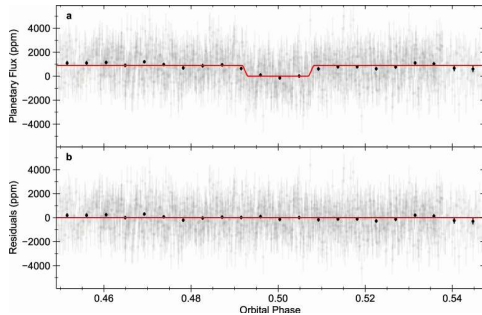
The  $4.05 \mu\text{m}$  absorption feature of  $\text{SO}_2$  in the JWST spectrum of the hot Saturn WASP-39 b ( $0.28 M_J$ ). Tsai et al. (2023).

**Photochemical**  $\text{SO}_2$  from UV photolysis of  $\text{H}_2\text{S}$ : the first photochemical product identified in an exoplanet atmosphere, proof of disequilibrium chemistry.

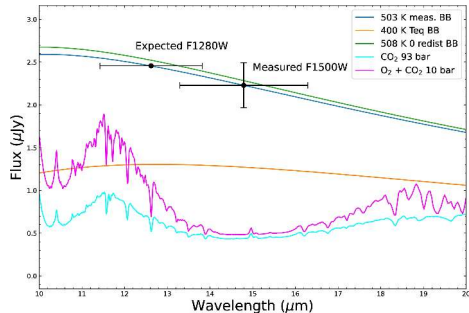


JWST/MIRI spectroscopic phase curve of WASP-43 b: thermal emission from dayside to nightside constrains the atmospheric heat redistribution. Credit: Bell et al. (2024).

# TRAPPIST-1 b: Bare Rock Dayside

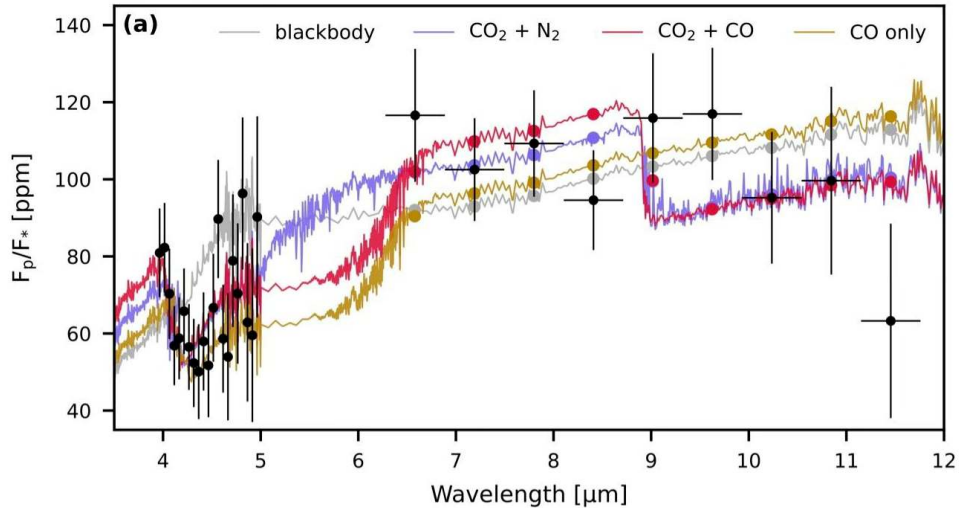


15  $\mu\text{m}$  secondary eclipse

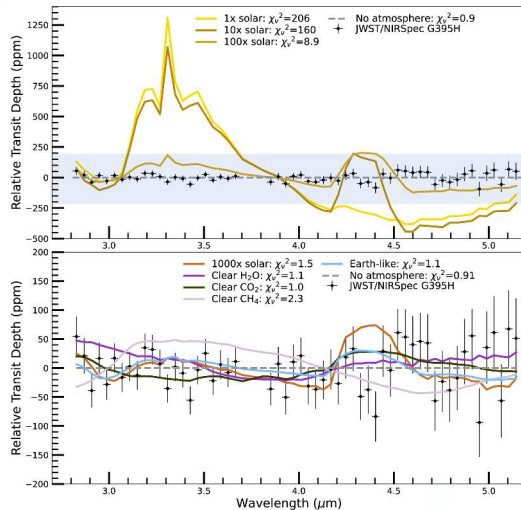


Compared to model atmospheres

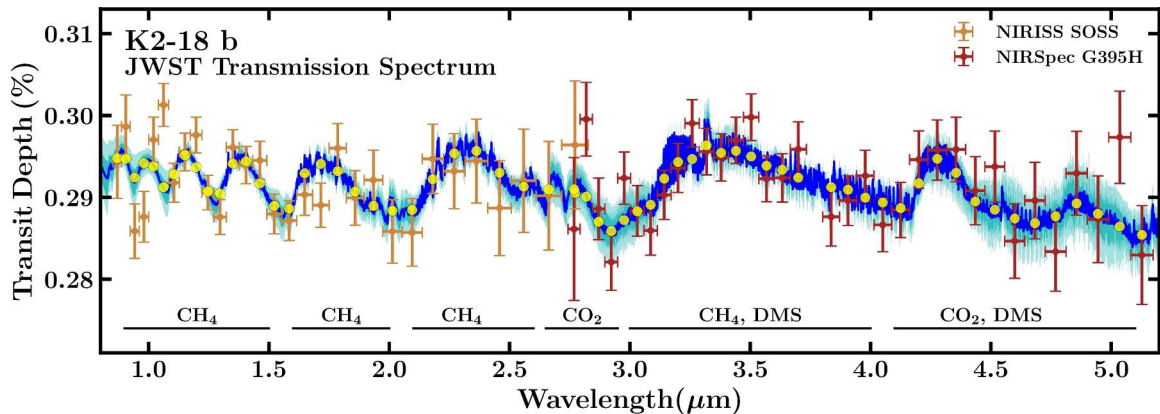
$T_B = 503^{+26}_{-27}$  K, consistent with bare rock; a **thick CO<sub>2</sub> atmosphere is ruled out** (Greene et al. 2023). With TRAPPIST-1 c and LHS 475 b, inner M dwarf rocky planets show a widespread lack of thick atmospheres.



JWST NIRCam and MIRI emission spectrum of 55 Cancri e: the data fall below the bare-rock blackbody and favour CO and  $\text{CO}_2$ , tentative evidence for a secondary atmosphere outgassed from a molten surface. Hu et al. (2024).

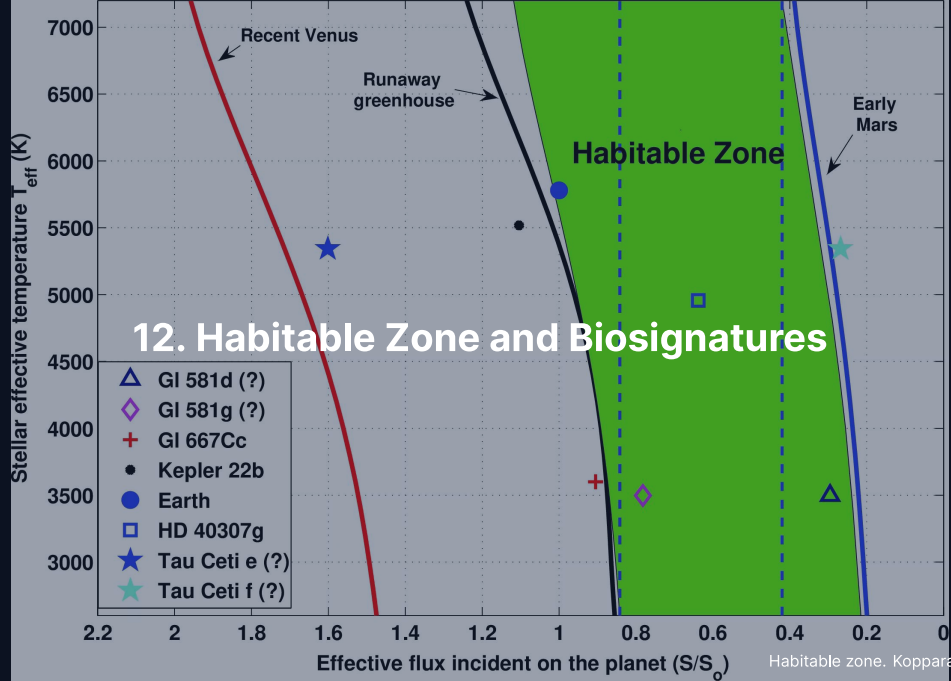


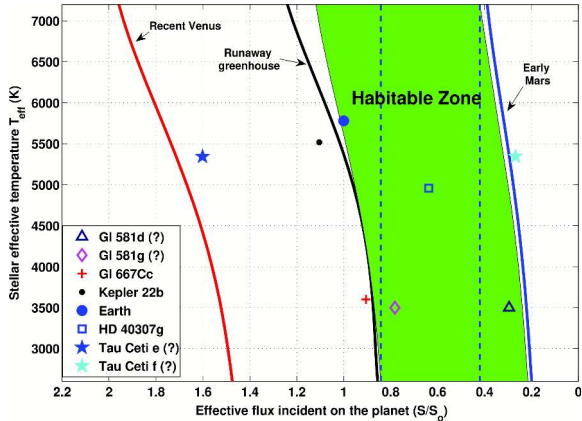
JWST NIRSpec G395H transmission spectrum of the Earth-size M dwarf planet LHS 475 b at about 12 pc: flat, so hydrogen-helium and pure  $\text{CH}_4$  atmospheres are ruled out and a Venus-like  $\text{CO}_2$  atmosphere stays marginally consistent. Lustig-Yaeger et al. (2023).



JWST transmission spectrum of the sub-Neptune K2-18 b in the habitable zone of its M dwarf. Madhusudhan et al. (2023).

$\text{CH}_4$  and  $\text{CO}_2$  detected, DMS disputed and tentative ( $\lesssim 2\sigma$ ); Hycean ocean, mini-Neptune envelope and magma ocean models compete: candidate biosignatures need rigorous real-time vetting.





The classical habitable zone from an updated 1D radiative-convective grid. Kopparapu et al. (2013).

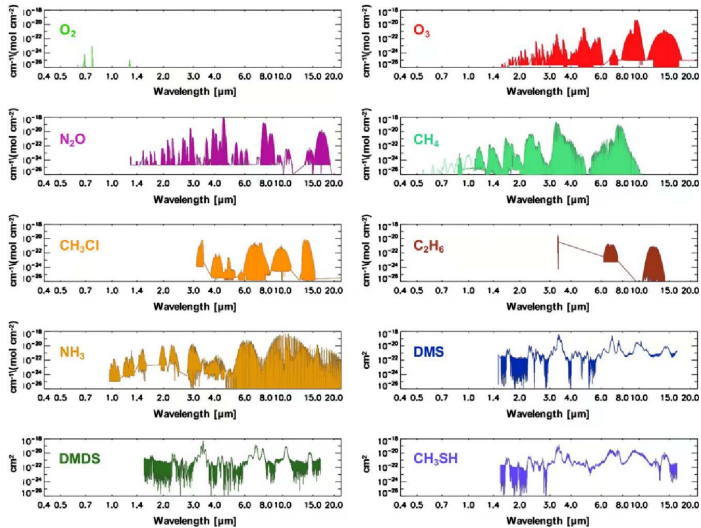
Inner edge: the Simpson-Nakajima runaway greenhouse; outer edge: the maximum  $\text{CO}_2$  greenhouse before  $\text{CO}_2$  condenses; both scale with stellar effective temperature and luminosity.

## Habitable Zone Evolution and Biosignatures

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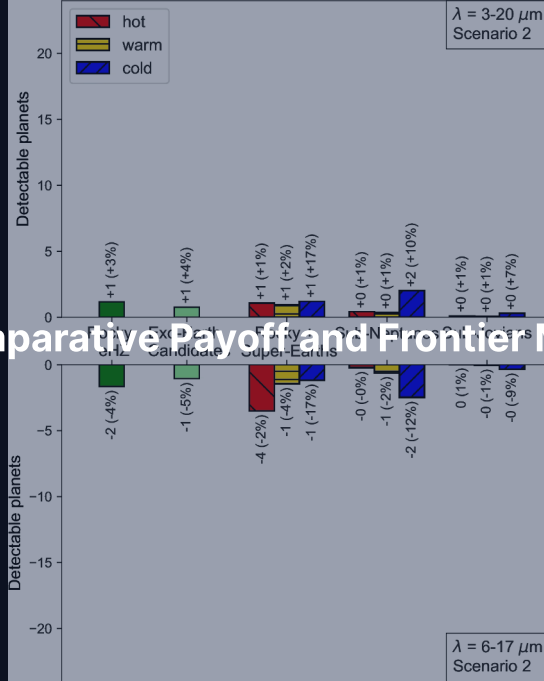
- **Dynamic trajectory:** HZ boundaries shift with stellar evolution; pre-MS M dwarfs subject planets to prolonged runaway desiccation
- **Disequilibrium pairs:** atmospheric biosignatures demand chemical disequilibrium ( $\text{O}_2 + \text{CH}_4$ ), not lone gas species
- **False positives:** abiotic  $\text{O}_2$  can build up by  $\text{H}_2\text{O}$  photolysis with hydrogen escape, or by  $\text{CO}_2$  photolysis

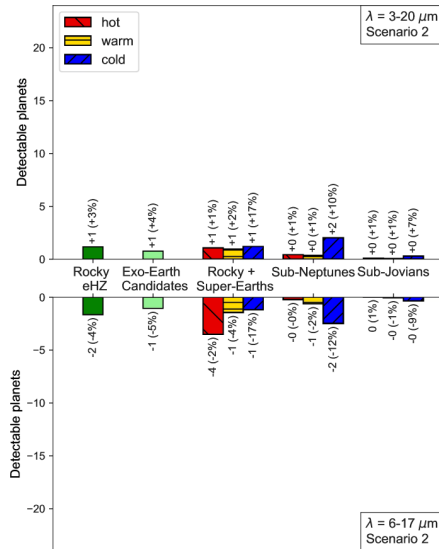
Life detection is an inverse problem requiring exclusion of all abiotic pathways.



Absorption cross-sections from 0.4 to 20  $\mu\text{m}$  for ten candidate biosignature gases: oxygen absorbs only in narrow optical bands, the others mostly in the infrared. Reproduced from Schwieterman et al. (2018).

# 13. Comparative Payoff and Frontier Missions





Predicted detection yields of the LIFE mid-infrared space interferometer for exoplanets within 20 pc, by radius and insolation. Quanz et al. (2022).

## Summary: Today's Course-Level Takeaways

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1. **Detection:** transit plus radial velocity breaks  $m \sin i$  and yields the true bulk density; every method carries its own bias
2. **Demographics:** small planets dominate ( $\eta_{\oplus} \sim 0.4$ ); photoevaporation or core-powered mass loss carves the radius valley at  $\sim 1.8 R_{\oplus}$ ; resonant chains and peas-in-a-pod systems record smooth disk migration
3. **Characterisation:** JWST detects photochemistry and bare rocks; biosignatures require ruling out false positives

**Next: Lecture 14**



# Questions?

## Next: Lecture 14. Synthesis & Astrobiology

Lecture notes: [formingworlds.github.io/IntroductionToPlanetaryScience](https://formingworlds.github.io/IntroductionToPlanetaryScience)